Only A Genius Can Solve Each Letter Is A Number_1080p.mp4

$$\begin{array}{cccccc} & S & E & N & D \\ + & M & O & R & E \\ \hline M & O & N & E & Y \end{array}$$

Mind Your Decisions

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Since the resulting sum is a 5 digit number, and $9999 + 9999 < 20000$, we can conclude $M = 1$.

Mind Your Decisions

$M = 1$

$$\begin{array}{cccccc} & & S & E & N & D \\ + & & 1 & O & R & E \\ \hline 1 & & O & N & E & Y \end{array}$$

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$M = 1$

$$\begin{array}{cccccc} & S & E & N & D \\ + & 1 & O & R & E \\ \hline 1 & O & N & E & Y \end{array}$$

Since $999 + 999 < 2000$, column 3 either gives 0 carryover, or it gives 1.

$$S + 1 = 10 + O \quad \Rightarrow S = 9 + O$$

$$1 + S + 1 = 10 + O \quad \Rightarrow S = 8 + O$$

Since S is a single digit, we must have $O = 0$ or $O = 1$.

Since $M = 1$, we must have $O = 0$.

Mind Your Decisions

$M = 1$

$O = 0$

$$\begin{array}{cccccc} & & S & E & N & D \\ + & & 1 & 0 & R & E \\ \hline 1 & 0 & N & E & Y \end{array}$$

Since $E \neq N$, we cannot have $E + 0 = N$. We must have $1 + E = N$. (There must be carryover from column 2).

Mind Your Decisions

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$M = 1$

$O = 0$

$$\begin{array}{cccccc} & S & E & N & D \\ + & 1 & 0 & R & E \\ \hline 1 & 0 & N & E & Y \end{array}$$

Since $E \neq N$, we cannot have $E + 0 = N$. We must have $1 + E = N$. (There must be carryover from column 2).

Since $N \neq 0$, we must have $E < 9$. Consequently, column 3 cannot produce any carryover.

Mind Your Decisions

$M = 1$

$O = 0$

$$\begin{array}{cccccc} & S & E & N & D \\ + & 1 & 0 & R & E \\ \hline 1 & 0 & N & E & Y \end{array}$$

Since $E \neq N$, we cannot have $E + 0 = N$. We must have $1 + E = N$. (There must be carryover from column 2).

Since $N \neq 0$, we must have $E < 9$. Consequently, column 3 cannot produce any carryover.

$$S + 1 = 10 \Longrightarrow S = 9$$

Mind Your Decisions

$M = 1$

$O = 0$

$S = 9$

$$\begin{array}{cccccc} & & 9 & E & N & D \\ + & & 1 & 0 & R & E \\ \hline 1 & 0 & N & E & Y \end{array}$$

Since $E \neq N$, we cannot have $E + 0 = N$. We must have $1 + E = N$. (There must be carryover from column 2).

Mind Your Decisions

$M = 1$

$O = 0$

$S = 9$

$$\begin{array}{cccccc} & & 9 & E & N & D \\ + & & 1 & 0 & R & E \\ \hline 1 & 0 & N & E & Y \end{array}$$

Since $E \neq N$, we cannot have $E + 0 = N$. We must have $1 + E = N$. (There must be carryover from column 2).

$$N + R = 10 + E$$

$$1 + N + R = 10 + E$$

Mind Your Decisions

$M = 1$

$O = 0$

$S = 9$

$$\begin{array}{cccccc} & & 9 & E & N & D \\ + & & 1 & 0 & R & E \\ \hline 1 & 0 & N & E & Y \end{array}$$

Since $E \neq N$, we cannot have $E + 0 = N$. We must have $1 + E = N$. (There must be carryover from column 2).

$$N + R = 10 + E \implies 1 + E + R = 10 + E \implies R = 9$$

$$1 + N + R = 10 + E$$

(But $R \neq 9$ since $S = 9$)

Mind Your Decisions

$M = 1$

$O = 0$

$S = 9$

$$\begin{array}{cccccc} & & 9 & E & N & D \\ + & & 1 & 0 & R & E \\ \hline 1 & 0 & N & E & Y \end{array}$$

Since $E \neq N$, we cannot have $E + 0 = N$. We must have $1 + E = N$. (There must be carryover from column 2).

$$1 + N + R = 10 + E \implies 1 + 1 + E + R = 10 + E$$

Mind Your Decisions

$M = 1$

$O = 0$

$S = 9$

$$\begin{array}{cccccc} & & 9 & E & N & D \\ + & & 1 & 0 & R & E \\ \hline 1 & 0 & N & E & Y & \end{array}$$

Since $E \neq N$, we cannot have $E + 0 = N$. We must have $1 + E = N$. (There must be carryover from column 2).

$$1 + N + R = 10 + E \implies 1 + 1 + E + R = 10 + E \implies R = 8$$

(Note: there must be carryover from column 1.)

Mind Your Decisions

$M = 1$

$O = 0$

$S = 9$

$R = 8$

$$\begin{array}{cccc} & 9 & E & N & D \\ + & 1 & 0 & 8 & E \\ \hline 1 & 0 & N & E & Y \end{array}$$

(Note: there must be carryover from column 1.)

Since $Y \neq 0$ or 1, we must have $D + E \geq 12$.

Since $D, E \neq 8$ or 9, we could have $7 + 5 = 12$ or $7 + 6 = 13$.

Mind Your Decisions

$M = 1$

$O = 0$

$S = 9$

$R = 8$

$$\begin{array}{cccccc} & & 9 & E & N & D \\ + & & 1 & 0 & 8 & E \\ \hline 1 & 0 & N & E & Y \end{array}$$

(Note: there must be carryover from column 1.)

Since $Y \neq 0$ or 1, we must have $D + E \geq 12$.

Since $D, E \neq 8$ or 9, we could have $7 + 5 = 12$ or $7 + 6 = 13$.

Either $E = 7$ or $D = 7$. But if $E = 7$, then $E + 1 = 8 = N$, which is not possible since $R = 8$.

Mind Your Decisions

$M = 1$

$O = 0$

$S = 9$

$R = 8$

$$\begin{array}{cccccc} & & 9 & E & N & D \\ + & & 1 & 0 & 8 & E \\ \hline 1 & 0 & N & E & Y \end{array}$$

(Note: there must be carryover from column 1.)

Since $Y \neq 0$ or 1, we must have $D + E \geq 12$.

Since $D, E \neq 8$ or 9, we could have $7 + 5 = 12$ or $7 + 6 = 13$.

Either $E = 7$ or $D = 7$. But if $E = 7$, then $E + 1 = 8 = N$, which is not possible since $R = 8$.

So $D = 7$, and $E = 5$ or 6.

Mind Your Decisions

$M = 1 \quad D = 7$

$O = 0$

$S = 9$

$R = 8$

$$\begin{array}{cccccc} & & 9 & E & N & 7 \\ + & & 1 & 0 & 8 & E \\ \hline 1 & 0 & N & E & Y \end{array}$$

$E = 5$ or 6.

If $E = 6$, then $E + 1 = 7 = N$, which is not possible since $D = 7$.

Hence $E = 5$, and $E + 1 = 6 = N$.

And finally $7 + E = 7 + 5 = 12$.

So $Y = 2$.

Mind Your Decisions

$M = 1 \quad D = 7$

$O = 0 \quad E = 5$

$S = 9 \quad N = 6$

$R = 8 \quad Y = 2$

$$\begin{array}{cccccc} & S & E & N & D \\ + & M & O & R & E \\ \hline M & O & N & E & Y \end{array}$$

$$\begin{array}{cccccc} & 9 & 5 & 6 & 7 \\ + & 1 & 0 & 8 & 5 \\ \hline 1 & 0 & 6 & 5 & 2 \end{array}$$

Mind Your Decisions